

DAIMON

Designing for Autonomy: AI Interactions,
Mental Outcomes, and Norms in Youth

Work Packages Structure

Instructions:

Please use Title, *Abstract* and **Role in the project** as *general reference*. Write the WP using the following format provided by the Horizon template. Each WP will be 2 pages long.

Table 3.1b: Work package description For each work package:

Work package number	
Work package title	

☐ Participants involved in each WP and their efforts are shown in table 3.1f. Lead participant and starting and end date of each WP are shown in table 3.1a.)

Objectives

Instructions, please remove

Description of work (where appropriate, broken down into tasks), lead partner and role of participants. For each task, quantify the amount of work. Provide enough detail to justify the resources requested and clarify why the work is needed and who will do it. Deliverables linked to each WP are listed in table 3.1c (no need to repeat the information here).

Introduction

Youth-facing AI systems are increasingly designed to minimize friction, optimizing engagement and retention rather than for the kind of effortful, resistant engagement that

sustains learning, critical thinking, and emotional resilience. DAIMON addresses this shift by treating epistemic friction as a measurable variable with direct consequences for cognitive autonomy and mental well-being in young users. The project moves from conceptual clarification, through technical operationalization, to large-scale empirical testing, and finally to policy-relevant deliverables. This trajectory is organized into eight Work Packages, each addressing a distinct stage of the research and innovation pathway, from the definition of foundational constructs to the dissemination of evidence-based guidelines for the design and regulation of AI companions targeting youth.

Overview of WPs and Participants

PI Fabio Grigenti (University of Padua)	WP1: Conceptual Foundations: Defining Epistemic Friction, Cognitive Autonomy, and Related Constructs
Marco Tamborini (Unipegaso)	WP2: From Concept to Artifact: Conceptual Analysis of AI Companions and Digital Tools
Bernardo Magnini (FBK – Fondazione Bruno Kessler)	WP3: Engineering the Epistemic Friction Score: Metric Design and LLM Behavioral Modeling
Antonella Marchetti (Università Cattolica Milan) – Davide Massaro (Università Cattolica Milan) – Mireia Vendrell Morancho (Artificial Intelligence research Institute - Spanish National Research Council)	WP4: Experimental Design Foundations: Linking Theoretical Constructs to Measurable Outcomes
Ioana Podina (Mind Care 4All)	WP5: The Dual-State RCT: Large-Scale Experimentation with Students
	WP6: Ecological Assessment of AI-Mediated Leisure Practices and Digital Well-being
PI Fabio Grigenti (University of Padua)	WP7: Project Management, Coordination, and Quality Assurance
PI Fabio Grigenti (University of Padua)	WP8: Dissemination, Communication, and Exploitation of Results

Work Packages (Section 3.1b)

WP1: Conceptual Foundations: Defining Epistemic Friction, Cognitive Autonomy, and Related Constructs

Abstract. This WP establishes the definitions for the project's core theoretical vocabulary. Drawing on epistemology, philosophy of education, and philosophy of mind, it distinguishes epistemic friction from mere difficulty or frustration and differentiates productive from unproductive forms of resistance. It further delineates the boundaries of Digital Cognitive Autonomy and Epistemic Agency as empirically testable constructs. The resulting conceptual scaffold anchors the vocabulary used by all subsequent technical and experimental Work Packages. Its output is a set of operative definitions suitable for translation into measurable variables.

Role in the project. WP1 supplies the foundational theoretical vocabulary on which the entire project depends; it is the upstream source from which the artifact-level categories of WP2 and the experimental metrics of WP4 are derived.

Objectives:

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Description of work:

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WP2: From Concept to Artifact: Conceptual Analysis of AI Companions and Digital Tools

Abstract. This WP analyses digital tools and AI companions not as neutral instruments but as practice-shaping artifacts embodying implicit pedagogical and cognitive theories. Through conceptual reverse-engineering, it reconstructs the design logics underlying anthropomorphic mirroring, instantaneous response, and frictionless compliance in current LLM-based systems. It compares these implicit theories against the constructs defined in WP1, producing a taxonomy of design patterns and their theorized cognitive and relational effects. This analysis is conducted at both a qualitative, theoretical level and a level oriented towards technical comparability. The output is a structured mapping between philosophical categories and artifact-level features.

Role in the project. WP2 functions as the conceptual hinge of the project, translating the abstract definitions produced in WP1 into artifact-specific categories that WP3 can operationalize into a computable score; without this mediating step, the philosophical and technical strands of the project would remain disconnected.

Objectives:

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Description of work:

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WP3: Engineering the Epistemic Friction Score: Metric Design and LLM Behavioral Modeling

Abstract. This WP develops the technical architecture of the Epistemic Friction Score, building internal probing frameworks to overcome the opacity of proprietary AI systems. It analyses how post-training interventions, including instruction tuning and related alignment procedures, shift a model's frictional behaviour along the dimensions identified in WP2. It defines calibrated, computable benchmarks that integrate quantitative performance with qualitative pedagogical adherence. It further develops concrete methodologies for deliberately increasing or decreasing epistemic friction in LLM-based systems. The output is a validated score together with a set of modification methodologies ready for experimental use.

Role in the project. WP3 provides the technical backbone of the project, transforming the conceptual categories of WP1 and WP2 into a reproducible, manipulable instrument that serves as the independent variable for the experimental trial conducted in WP5.

Objectives:

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Description of work:

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WP4: Experimental Design Foundations: Linking Theoretical Constructs to Measurable Outcomes

Abstract. This WP defines the theoretical and methodological architecture underlying the social experiment. It operationalizes constructs such as cognitive autonomy, epistemic agency, and relevant mental health indicators into measurable instruments, including the Trust Trap protocol and associated psychometric scales. It specifies the correlational structure linking the Epistemic Friction Score to educational, cognitive, and emotional outcomes. It establishes the hypotheses, sampling strategy, and measurement plan that will govern the trial. The output is a validated experimental protocol ready for deployment at scale.

Role in the project. WP4 acts as the methodological bridge between the technical score produced in WP3 and the empirical trial conducted in WP5, ensuring that the large-scale experiment rests on validated constructs and instruments rather than on ad hoc measures.

Objectives:

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Description of work:

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WP5: The Dual-State RCT: Large-Scale Experimentation with Students

Abstract. This WP implements a controlled trial involving 2000 students selected across age groups, socio-economic backgrounds, cultural contexts, and ability profiles. It deploys AI systems calibrated to different Epistemic Friction Score values together with the Trust Trap protocol designed in WP4. It collects empirical data on cognitive focus, learning performance, epistemic agency, and emotional and mental health indicators. It tests the project's central hypothesis under realistic interaction conditions with youth. The output is the primary empirical dataset of the project.

Role in the project. WP5 constitutes the empirical core of DAIMON, the point at which the theoretical hypotheses and calibrated instruments developed in WP1 through WP4 are tested against real interactions between young users and AI systems, generating the evidence base for all downstream deliverables.

Objectives:

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Description of work:

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WP6: Ecological Assessment of AI-Mediated Leisure Practices and Digital Well-being

Abstract. This WP investigates the ecological dimension of AI-mediated leisure practices among youth. It employs longitudinal and Experience Sampling Method (ESM) designs to observe everyday interactions with digital companions outside of educational settings. It maps current media repertoires and tracks daily fluctuations in digital well-being, emotional regulation, and perceived loneliness. It specifies the contextual variables mediating algorithmic passivity versus active engagement. The output is a comprehensive empirical dataset detailing the real-world impact of frictionless AI usage on adolescent cognitive and emotional habits.

Role in the project. This WP acts as the anchor between the theoretical framework and the controlled experimental phases, ensuring that the project's hypotheses and technical metrics remain firmly grounded in the actual media routines and daily digital realities of European youth.

Objectives:

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Description of work:

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WP7: Project Management, Coordination, and Quality Assurance

Abstract. This WP oversees the overall governance of the consortium, including administrative, financial, and legal compliance with Horizon Europe requirements. It coordinates collaboration across philosophy, psychology, AI, NGO, and industrial partners involved in the project. It manages risk, timeline adherence, and internal quality assurance across all Work Packages. It ensures compliance with ethical requirements arising from the involvement of minors as research participants. The output is the sustained, coherent execution of the project across its full duration.

Role in the project. WP7 guarantees the coherent, timely, and ethically compliant execution and integration of the seven research and innovation Work Packages, acting as the connective management layer of the consortium.

Objectives:

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Description of work:

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WP8: Dissemination, Communication, and Exploitation of Results

Abstract. This WP plans and implements the dissemination of DAIMON's findings to academic, policy, and public audiences. It organizes workshops and scientific publications communicating the project's conceptual, technical, and empirical results. It engages schools, parents, regulators, and industry stakeholders with the guidelines and policy framework produced in WP6. It supports exploitation pathways for partner organizations, including the industrial partner. The output is a structured communication and exploitation plan ensuring uptake beyond the project's lifetime.

Role in the project. WP8 amplifies and operationalizes the societal and policy reach of the project's outputs, ensuring that the evidence and guidelines generated by WP1 through WP6 achieve sustained uptake among policymakers, educators, and the AI industry.

Objectives:

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Description of work:

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